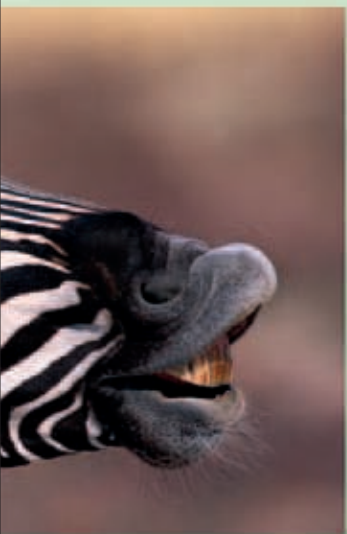




Something in the air

Curling back its upper lip, this male zebra is testing the air around the hind quarters of a female for pheromones. This behavior, called flehmen, passes air over the Jacobson's organ, which is in the roof of its mouth (see p. 120). This organ detects pheromones in tiny amounts, letting the zebra know if the female is ready to breed.



Smells familiar

Minutes after being born on Africa's grassy plains, a young Thomson's gazelle is helped to its feet by its mother. As she licks her calf, she memorizes its scent, helping her to identify it until it is old enough to fend for itself. From this moment onward, she feeds her calf on her milk, but will push away any "foreign-smelling" calves.

Secret signals

Insects use pheromones throughout their lives. Some kinds—such as alarm pheromones—produce sudden changes in behavior. Others work more slowly, affecting the way insects grow and develop. Queen bees release slow-acting pheromones to control their workers and to keep their nests running smoothly.



Sex pheromones

Powerful attractants called sex pheromones bring males and females together so that they can mate. In grasshoppers—and most other insects—the female produces the pheromone, guiding the male toward her.



Spacing pheromones

Like many other insects, plant bugs leave spacing pheromones when they lay their eggs. These pheromones "tag" plants that have eggs, so that other females do not lay theirs in the same place.



Trail pheromones

Processionary moth caterpillars live in silk nests in trees. When they emerge to feed, they move in formation, laying down a trail pheromone. This later guides them back to their nest.



Alarm pheromones

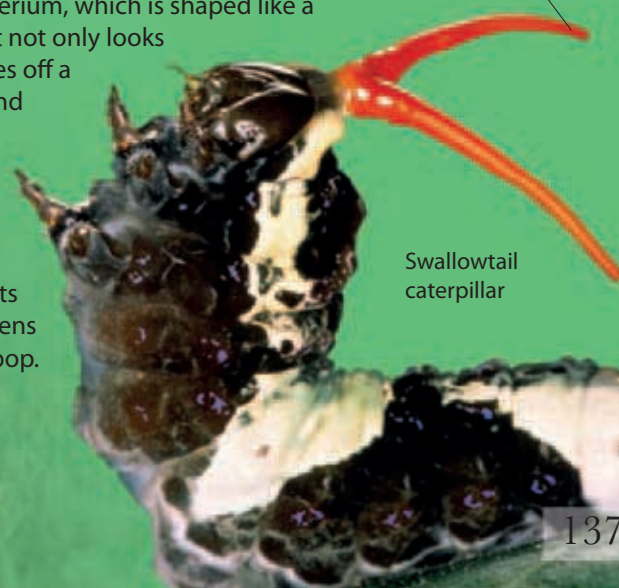
If honeybees are attacked or injured, they release an alarm pheromone, which makes other bees go on the attack. These bees also give off the pheromone, so thousands soon join in.

Stink bombs

In addition to making pheromones, many insects use chemicals for self-defense. Threatened by a predator, this swallowtail caterpillar has inflated a gland called an osmeterium, which is shaped like a pair of miniature horns. It not only looks alarming, but it also gives off a powerful smell. Beetles and plant-sucking bugs are also expert at this kind of chemical warfare.

Bombardier beetles produce an evil-smelling explosive spray that squirts out of their abdomens with a loud pop.

Osmeterium extends to form a pair of horn-shaped scent weapons



Swallowtail caterpillar